

Maintenance History Log

Equipment P-101 — Crude Charge Pump — Extract covering 2024 to 2026

SYNTHETIC DOCUMENT - created for a Smart India Hackathon prototype demonstration. Not real plant data. No sponsor information is used.

1. Work Orders Raised

WO number	Date	Description	Downtime (h)
WO-24118	2024-04-12	Mechanical seal replaced, routine	14
WO-24406	2024-10-16	Mechanical seal replaced, leakage	18
WO-25077	2025-03-02	Bearing replaced, high vibration	22
WO-25214	2025-06-19	Mechanical seal replaced, leakage	16
WO-25390	2025-09-21	Seal flush strainer cleaned	4
WO-25502	2025-12-24	Mechanical seal replaced, leakage	19
WO-26041	2026-03-28	Mechanical seal replaced, leakage	17
WO-26129	2026-06-30	Coupling alignment check	6

2. Seal Failure Intervals

Failure	Date	Days since previous	Note
1	2024-04-12	-	Routine replacement
2	2024-10-16	187	Leakage observed at gland
3	2025-06-19	246	Leakage, flush dP not recorded
4	2025-12-24	188	Leakage, strainer found fouled
5	2026-03-28	94	Leakage, interval halved

The interval between seal failures has reduced from 187 days to 94 days over the period covered. Flush line differential pressure was not recorded consistently before September 2025.

3. Consumables Used

Part number	Description	Quantity	Unit cost (Rs.)
SEAL-682-C2	Cartridge mechanical seal	5	184000
FLT-25M-04	Flush filter element 25 micron	9	3200
BRG-7312-BG	Angular contact bearing	2	26500